Einstein's Heuristic as Statistical Inference: The Photoelectric Effect as Machine Learning, 1887–1916

Thomas J. Sargent*

March 2026

Abstract

In his 1905 "miraculous year" paper on the photoelectric effect, Albert Einstein proposed that electromagnetic radiation consists of discrete energy quanta $h\nu$. His argument had two components: a physical *model* (the light-quantum hypothesis) and an implied *measurement equation* relating the observable stopping potential V_0 linearly to the incident frequency ν. This note examines Einstein's reasoning from a modern statistical perspective. Setting Millikan's measurements against the qualitative findings of Lenard (1902) and the discrepant estimates of Hughes (1912) and Richardson & Compton (1912), I carry out the ordinary least-squares regression that the theory implies: $V_0 = \alpha + \beta\nu + \varepsilon$. The slope estimate $\hat{\beta}$ estimates h/e, and hence Planck's constant h. Millikan's five-point sodium regression gives $\hat{\beta} = 4.128 \times 10^{-15}$ V·s, which with the modern elementary charge implies $h = 6.62 \times 10^{-34}$ J·s, 0.17% below the defined value; the larger error in Millikan's own published h is inherited from his oil-drop charge, not from the photoelectric slope. A maximum likelihood treatment under Gaussian errors recovers the same point estimates and shows that OLS attains the Cramér-Rao bound. The tabulated stopping potentials are reconstructions that lie on a fitted line to within a fraction of a millivolt, so their residuals cannot furnish a standard

*This paper is one of five that read episodes in the history of physics and astronomy as exercises in statistical inference and machine learning. Sargent [2025] sets out the theme: what is now called artificial intelligence names practices that scientists have used for centuries. The other four treat Kepler and Newton, Pauli's exclusion principle, the exoplanet discovery of Mayor and Queloz, and the accelerating universe of Riess, Schmidt and Perlmutter.

error. I take the uncertainty from Millikan's nine replicated two-point sessions (0.7%). It agrees with the Cramér-Rao bound at his instrumental precision (0.8%). That bound also explains his central design choice. The standard error on h/e falls with the spread of the frequencies, not with the number of readings. Millikan's measured slope also agrees to 0.35% with the slope that Planck's 1901 blackbody constant and Millikan's own elementary charge predict, a comparison in which no photoelectric datum appears. I then read the episode in the language of machine learning. Einstein's quantum hypothesis supplied the hypothesis class and the inductive bias. Lenard's feature-importance finding eliminated intensity as an input. The regression is a supervised-learning problem in which five training points suffice because the physics fixed the functional form. A deep network given the same data recovers neither the constant nor its interpretation. The regression selects a functional form and not a mechanism: Einstein's equation follows without quantising the electromagnetic field, and photon-counting experiments settled that question only in 1977. Theory reduces the sample size that data must supply, and it also determines which measurement discriminates.

1 Introduction

In December 1900, Max Planck introduced the energy quantum $h\nu$ as "an act of desperation" to fit the blackbody spectrum. Five years later, Einstein borrowed that quantum and applied it to a second phenomenon—the photoelectric effect—in a paper that would earn him the 1921 Nobel Prize. Einstein posited a falsifiable linear equation relating the stopping potential of emitted electrons to the incident frequency. Its slope is h/e. Millikan confirmed the equation to better than 1% in 1916.

The 1905 paper was also a statistical argument, though not the one later readers often ascribe to it. Einstein's route to the light quantum was thermodynamic. He showed that in the Wien region the *entropy* of radiation depends on volume as the entropy of an ideal gas of n independent particles does. He inferred that radiation behaves, for these purposes, as a collection of independent localised quanta. He did not in 1905 read the classical intensity $|E|^2$ as a probability density for locating a quantum. He took that step in the 1920s, in the guise of the *Gespensterfeld* or ghost field: a wave that guides probabilities without itself carrying energy. Max Born made that reading the cornerstone of quantum mechanics in 1926 [Born, 1926], crediting Einstein's guiding field as he generalised it from photons to the Schrödinger wave function. Einstein won the 1921 Nobel Prize for the photoelectric equation. Born won the 1954 Prize for the probabilistic interpretation. Einstein spent the rest of his life opposing it.

A modern data scientist confronted with a table of (frequency, voltage) pairs would recognise the photoelectric experiment as a two-variable linear regression problem. The regression came after the theoretical insight. Albert Einstein in 1905 wrote down the model

$$V_0 = \frac{h}{e}\,\nu - \frac{W}{e} \tag{1}$$

before anyone had measured a reliable V_0–ν slope. Equation (1) is a prediction from what Planck himself called "simply an act of desperation"[1]: that light arrives in discrete quanta of energy $h\nu$. The slope coefficient is h/e, where h is Planck's constant and e the elementary charge. The intercept is $-W/e$, where W is the metal's work function—the energy cost of extracting an electron from the bulk.

Eleven years later, Robert Millikan [Millikan, 1916] provided the definitive empirical confirmation, using freshly scraped alkali-metal surfaces in high vacuum and narrow-band filtered spectral lines from a mercury arc. His data, assembled in Table 3 below, allow a textbook OLS regression that recovers h to within 1% of its modern value.

The episode provides a microcosm of scientific inference:

1. **Model specification.** Einstein's quantum hypothesis imposes linearity of V_0 in ν, a slope h/e that is the same for every metal, and a metal-specific intercept.

2. **Experimental design as data quality control.** Contaminated surfaces, stray light, and a restricted frequency range biased the estimates of Lenard, Hughes, and Richardson & Compton. Millikan solved these problems one by one.

3. **Parameter identification.** Variation in ν across widely separated spectral lines identifies the slope h/e. The intercept identifies the work function W only up to the contact electromotive force (EMF) between the photoemitter and the collector, a nuisance parameter that shifts the intercept but not the slope.

4. **Reluctance to accept the model.** Millikan confirmed Einstein's equation while

[1] Planck introduced the energy quantum $h\nu$ in December 1900 to derive the blackbody radiation spectrum, later describing his own step in a letter to R. W. Wood (7 October 1931): "Briefly summarized, what I did can be described as simply an act of desperation... a theoretical interpretation had to be found at any cost, no matter how high." [Planck, 1931]. Einstein borrowed the quantum $h\nu$ five years later and applied it to photoemission.

publicly rejecting the light-quantum interpretation.

5. **Unintended conceptual consequences.** The picture that motivated the equation—a field that fixes probabilities for discrete quanta—became, twenty years later, Born's rule [Born, 1926].

Section 2 sketches the experimental history. Section 3 develops the regression framework. Section 4 assembles the historical data. Section 5 presents the OLS estimates and diagnostics. Section 6 develops maximum likelihood estimation of Einstein's parameters and derives the Cramér-Rao bound. Section 7 discusses the contact-EMF identification problem. Section 8 analyses Millikan's nine two-point determinations. Section 9 compares historical estimates of h. Section 10 distinguishes inference from interpretation. Section 11 reinterprets the episode in the language of machine learning and AI. Section 12 contrasts Einstein's regression with what a deep neural network would learn from the data available at each epoch. Section 13 concludes.

Appendix A.4 shows that Einstein's equation follows without quantising the electromagnetic field, so Millikan's regression selects a functional form and not a mechanism. Appendix A.5 describes the photon-counting experiments that settled the mechanism in 1977. Appendix A.6 draws the consequence for Section 11. Theory reduces the sample size that data must supply, and theory also determines which measurement discriminates.

The same statistical anatomy governs the discovery of cosmic acceleration by Riess, Schmidt and Perlmutter in 1998, which a companion paper treats in the same way. General relativity replaces the quantum hypothesis in fixing the functional form. A combination of the Hubble constant and the supernovae's absolute magnitude replaces the contact EMF as the nuisance parameter that shifts the level of the diagram and leaves its shape alone. And the fit again settles a functional form without settling the physics behind it. The difference is that Millikan's slope is a single number, fully identified once the nuisance parameter is removed, while the Friedmann model has two shape parameters that remain tangled with each other. The photoelectric case is the one in which identification succeeds completely, and reading it that way makes the harder case legible.

2 Experimental History, 1887–1916

2.1 Discovery and Qualitative Characterisation

Heinrich Hertz observed in 1887 that ultraviolet light reduced the sparking distance in a gap discharge. Wilhelm Hallwachs (1888) showed that a zinc plate illuminated by UV light lost negative charge. These observations established that light could eject negatively charged particles from metals.

J. J. Thomson (1899) identified the ejected particles as electrons by measuring their charge-to-mass ratio. The same year Philipp Lenard began systematic experiments, culminating in his landmark 1902 paper [Lenard, 1902].

2.2 Lenard's Key Finding (1902)

Lenard's decisive contribution was establishing two qualitative facts:

1. The *maximum kinetic energy* (equivalently the stopping potential V_0) of the emitted photoelectrons is **independent of light intensity**. Classical wave theory predicted the opposite: more intense waves deliver more energy.
2. V_0 depends on the **frequency** ("colour") of the incident light, with higher frequency producing higher V_0.

Lenard's quantitative data were, by his own account, rough: his metal surfaces were oxidised, vacuum conditions poor, and no attempt was made to separate the dependence of V_0 on ν from confounding effects. He received the 1905 Nobel Prize in Physics for his cathode-ray work.

2.3 Einstein's 1905 Hypothesis

In March 1905 [Einstein, 1905] Einstein proposed that each light quantum carries energy $h\nu$. A single quantum is absorbed by a single electron. The electron escapes if $h\nu > W$; the excess appears as kinetic energy, so the stopping potential satisfies equation (1). At the time of writing, Einstein noted:

> "As far as I can see, our conception is not at variance with the properties of the photoelectric effect observed by Mr. Lenard."

There were, however, no quantitative measurements of the V_0–ν slope. The predicted slope h/e is a precise, falsifiable number. Since the 2019 redefinition of the SI both h and e are exact, so the target is exact too: $h/e = 4.135667 \times 10^{-15}$ V·s. In 1905 the number was not known to that accuracy, but it was already fixed—by Planck's blackbody fit five years earlier, with no freedom to adjust it to photoelectric data.

2.4 Early Experiments (1907–1914)

Between 1907 and 1914 three groups attempted quantitative verification. Table 6 summarises the estimates of h from the two that published numbers precise enough to enter it.

Ladenburg (1907). The first quantitative V_0–ν measurements, but confined to a wavelength range of some 60 nm (2000–2600 Å). Over so narrow a span a linear and a square-root law are nearly indistinguishable, and Ladenburg's data could not discriminate between Einstein's hypothesis and its classical rivals. He reported no determination of h stable enough to tabulate, and none appears in Table 6.

Hughes (1912). Arthur Llewelyn Hughes [Hughes, 1913] used three UV lines ($\lambda =$ 1849, 2257, 2537 Å) and roughly a dozen metals. Two of the three frequencies were used to fit the line; the third served as a check. Hughes reported h values ranging from 3.55×10^{-27} to 5.85×10^{-27} erg·s—a spread of 65%. Hughes did not measure or correct the contact potentials. The 69 nm span of his three lines is too narrow to pin down a slope. Millikan's six lines span 293 nm.

Richardson & Compton (1912). Owen Richardson and Karl Compton [Richardson and Compton, 1912] measured eight metals. Their h estimates varied by more than 60%. Millikan [1916] attributed the scatter to the same deficiencies as Hughes: narrow frequency range, uncontrolled contact EMF, and stray short-wavelength scattered light.

2.5 Millikan's Definitive Measurement (1916)

Robert Millikan spent a decade developing his "machine shop in vacuo." Inside an evacuated glass bulb, an electromagnetically driven knife freshly shaved the surface of a cast cylinder of sodium, potassium, or lithium. The wheel rotated the fresh surface into the light beam without breaking vacuum. The contact EMF was independently measured by the Kelvin double-balance method and monitored throughout each run. Only spectral lines with no short-wavelength companions in the mercury arc were used, and those lines were further filtered to remove scattered UV.

The result was the regression in Section 5.

3 The Regression Equation

3.1 Derivation

Einstein's light-quantum hypothesis and the energy-conservation condition for photoelectron escape give

$$\underbrace{eV_0}_{\text{max KE}} = \underbrace{h\nu}_{\text{photon energy}} - \underbrace{W}_{\text{work function}}, \tag{2}$$

where all quantities are in SI units. Dividing by e gives the true stopping potential as a function of frequency,

$$V_0 = \underbrace{\frac{h}{e}}_{\beta}\,\nu \underbrace{-\frac{W}{e}}_{\text{work function}} + \underbrace{\varepsilon}_{\text{meas. error}}. \tag{3}$$

The apparatus reads an applied potential displaced by the contact EMF v_c between photoemitter and collector,

$$V_0^{\text{obs}} = V_0 - v_c = \frac{h}{e}\,\nu - \frac{W}{e} - v_c + \varepsilon. \tag{4}$$

Because v_c enters additively it shifts the intercept and leaves the slope alone. Writing α for whichever intercept is in force, the estimating equation is

$$\boxed{V_0 = \alpha + \beta\,\nu + \varepsilon, \qquad \beta = \frac{h}{e}.} \tag{5}$$

This is a simple bivariate linear regression with one unknown slope (β) and one unknown intercept (α).

Convention. Millikan measured v_c independently by the Kelvin method and added it back before plotting, and the stopping potentials tabulated in Section 4 are the corrected values $V_0 = V_0^{\text{obs}} + v_c$. Every regression reported below is run on those corrected values, so throughout this paper

$$\alpha = -W/e, \tag{6}$$

and the work function is recovered from the intercept alone, $W/e = -\hat{\alpha}$. Had the raw readings been used instead, the intercept would have been $-W/e - v_c$, and separating the two would have required an independent measurement of v_c. Adding v_c to $-\hat{\alpha}$ when the data already contain the correction double-counts the contact EMF.

3.2 Identification

The slope. Variation in ν across spectral lines identifies $\beta = h/e$. The standard error of $\hat{\beta}$ is $s/\sqrt{S_{\nu\nu}}$, where $S_{\nu\nu} = \sum_i(\nu_i - \bar{\nu})^2$; spreading the ν_i apart raises $S_{\nu\nu}$ and shrinks the standard error. The early experiments were confined to UV bands some 60–70 nm wide and gave unreliable estimates; Section 6.5 quantifies the penalty.

The intercept and the contact-EMF problem. The intercept is contaminated by the contact EMF v_c, which was the dominant experimental nuisance of the whole literature: it could reach 3 V, comparable to the stopping potentials themselves, and it drifted with surface age, adsorbed gas, and temperature. Because it enters additively it is harmless to the slope so long as it is constant within a run, and fatal to it otherwise. Section 7 takes up both the identification problem and Millikan's solution to it.

The universal slope as a cross-metal restriction. Because $\beta = h/e$ is a universal physical constant, equation (5) implies that *all* metals share the same slope. This is a strong and testable restriction. Economists will recognise a cross-equation restriction, in which a single deep parameter appears identically in the reduced form of every metal. Separate regressions must return the same $\hat{\beta}$. Millikan reported that his sodium and lithium slopes did. Section 6.6 imposes the restriction, tests it, and explains why his data support it less strongly than the raw agreement suggests.

4 The Historical Data

4.1 Millikan's Mercury Arc Lines

Millikan used isolated lines from a mercury arc whose wavelengths were independently measured by Walter T. Whitney by grating photograph. Table 1 lists the lines and their frequencies (computed from $\nu = c/\lambda$ with $c = 2.998 \times 10^{10}$ cm/s).

Table 1: Mercury-arc spectral lines used by Millikan [1916]. Frequencies computed from $\nu = c/\lambda$, $c = 2.998 \times 10^{10}$ cm/s.

λ (Å)	λ (nm)	ν (10^{14} Hz)
5461.0	546.10	5.490
4339.4	433.94	6.908
4046.8	404.68	7.409
3650.2	365.02	8.213
3125.5	312.55	9.591
2534.7	253.47	11.827

4.2 Millikan's Raw Stopping-Potential Data

Table 2 reproduces the control readings near the balance point (zero electrometer deflection) from Millikan's Table I. The electrometer sensitivity was 2500 mm/V; the Weston voltmeter was certified by the National Bureau of Standards to ±0.1%.

Table 2: Selected readings from Millikan (1916), Table I. Sodium surface. "V" is the applied potentiometer voltage; "mm" is the electrometer deflection in 30 s. For lines 5461–3126 Å the contact EMF ($v_c = 2.51$ V) exceeds the photopotential and the reported voltages are the *assisting* potential near balance (balance = zero deflection). For line 2535 Å voltages are retarding.

5461 Å		4339 Å		4047 Å		3650 Å		3126 Å		2535 Å	
V	mm	V	mm	V	mm	V	mm	V	mm	V	mm
2.257	28	1.581	44	1.576	82	1.157	67.5	0.581	52	−0.058	68
2.205	14	1.629	20	1.524	55	1.105	36	0.529	29	+0.058	38
2.152	7	1.576	10	1.471	36	1.053	19	0.477	12	+0.162	26
2.100	3	1.524	4	1.419	24	1.000	11	0.424	5	+0.267	16.5
		1.367	3	0.948	4			0.372	2.5	+0.372	8

The balance points interpolated from these curves (Millikan's Fig. 5) are the stopping potentials used in the regression.

4.3 Reconstructed Stopping Potentials

Table 3 collects the true stopping potentials $V_{0,\text{true}} = V_{0,\text{obs}} + v_c$ for sodium and lithium. For sodium, the contact EMF was independently measured as $v_c = 2.51$ V (stable to

± 0.01 V over 9 hours). For lithium (first series), $v_c = 1.52$ V. For sodium at 2535 Å the paper explicitly states $V_{0,\text{obs}} = 0.52$ V, giving $V_{0,\text{true}} = 3.03$ V, which lies 0.04 V below the value implied by the line fitted to the other five points.

Provenance of the tabulated values. The 2535 Å sodium entry is Millikan's printed figure. The other five sodium entries are reconstructions. I read them off the straight line Millikan fitted to his Fig. 5 balance points and rounded them to three decimals. His plotted curves cannot be read to better than a few hundredths of a volt. The lithium column is a reconstruction too. Its entries at 4339–3126 Å are the sodium entries less a constant 0.540 V.

Two consequences follow. Residuals computed from these five points measure rounding, not measurement scatter, and a standard error built from them understates the uncertainty by more than an order of magnitude. The sodium and lithium slopes agree over the four shared lines by construction.

Sections 5 and 6 therefore exhibit the inferential logic that Einstein's equation implies. Section 8 supplies the uncertainty estimate. Millikan's nine two-point sessions are repeated measurements with fresh surfaces.

Table 3: Historical stopping-potential data used in the regression. True stopping potentials $V_0 = V_{0,\text{obs}} + v_c$ for sodium (Na, $v_c = 2.51$ V) and lithium (Li, $v_c = 1.52$ V). These values are reconstructed from Millikan (1916) as described in Section 4; only the Na entry at 2535 Å is a directly quoted figure. The Li entries at 4339–3126 Å are the Na entries less a constant 0.540 V, so the two slopes agree over those lines by construction (Section 4). The dash for Li at 5461 Å indicates that this wavelength is above lithium's photoemission threshold ($\lambda_0 \approx 5260$ Å).

λ (Å)	λ (nm)	ν (10^{14} Hz)	V_0^{Na} (V)	V_0^{Li} (V)
5461.0	546.10	5.490	0.454	—
4339.4	433.94	6.908	1.039	0.499
4046.8	404.68	7.409	1.246	0.706
3650.2	365.02	8.213	1.578	1.037
3125.5	312.55	9.591	2.147	1.606
2534.7	253.47	11.827	3.030	2.529

5 OLS Regression: Estimating h/e

5.1 Regression Setup

The model is

$$V_{0i} = \alpha + \beta\,\nu_i + \varepsilon_i, \quad \varepsilon_i \overset{\text{iid}}{\sim} (0, \sigma^2), \quad i = 1, \ldots, n.$$

OLS minimises $\sum_{i=1}^{n}(V_{0i} - \alpha - \beta\nu_i)^2$. The closed-form estimates are

$$\hat{\beta} = \frac{S_{\nu V}}{S_{\nu\nu}} = \frac{\sum_i(\nu_i - \bar{\nu})(V_{0i} - \bar{V}_0)}{\sum_i(\nu_i - \bar{\nu})^2}, \tag{7}$$

$$\hat{\alpha} = \bar{V}_0 - \hat{\beta}\,\bar{\nu}. \tag{8}$$

5.2 Results: Sodium (Five Points, $\lambda = 5461$–3126 Å)

Millikan's primary regression used the five most reliable sodium points (omitting 2535 Å where a scattered-light correction was required). I reproduce this regression exactly.

Data.

ν (10^{14} Hz)	5.490	6.908	7.409	8.213	9.591
V_0 (V)	0.454	1.039	1.246	1.578	2.147

Summary statistics.

$$\begin{aligned} n &= 5, \\ \bar{\nu} &= \tfrac{1}{5}(5.490 + 6.908 + 7.409 + 8.213 + 9.591) = 7.522 \times 10^{14}\ \text{Hz}, \\ \bar{V}_0 &= \tfrac{1}{5}(0.454 + 1.039 + 1.246 + 1.578 + 2.147) = 1.293\ \text{V}. \end{aligned}$$

Sums of squares and cross-products.

i	$\nu_i - \bar{\nu}$ (10^{14} Hz)	$V_{0i} - \bar{V}_0$ (V)	$(\nu_i - \bar{\nu})^2$	$(\nu_i - \bar{\nu})(V_{0i} - \bar{V}_0)$
1	−2.032	−0.839	4.129	1.705
2	−0.614	−0.254	0.377	0.156
3	−0.113	−0.047	0.013	0.005
4	+0.691	+0.285	0.477	0.197
5	+2.069	+0.854	4.281	1.767
$\sum$			$S_{\nu\nu} = 9.277$	$S_{\nu V} = 3.830$

(Units: $S_{\nu\nu}$ in $(10^{14}\text{ Hz})^2$; $S_{\nu V}$ in 10^{14} Hz·V.)

OLS estimates.

$$\hat{\beta} = \frac{S_{\nu V}}{S_{\nu\nu}} = \frac{3.830}{9.277} = \mathbf{0.41285} \times 10^{-14} \text{ V/Hz} = \mathbf{4.128 \times 10^{-15}} \text{ V·s}, \tag{9}$$

$$\hat{\alpha} = \bar{V}_0 - \hat{\beta}\,\bar{\nu} = 1.293 - 4.128 \times 10^{-15} \times 7.522 \times 10^{14} = 1.293 - 3.106 = -\mathbf{1.813} \text{ V}. \tag{10}$$

Residuals and goodness of fit.

λ (Å)	V_{0i} (V)	$\hat{V}_{0i}$ (V)	$\hat{\varepsilon}_i$ (mV)
5461	0.454	0.4538	+0.18
4339	1.039	1.0392	−0.23
4047	1.246	1.2461	−0.07
3650	1.578	1.5780	+0.01
3126	2.147	2.1469	+0.11

The coefficient of determination is $R^2 = 1 - \text{RSS}/S_{VV} = 1 - (1.03 \times 10^{-7})/(1.58) \approx 0.9999999$.

The residuals measure rounding. Every residual is smaller than 0.25 mV, forty times below Millikan's stated reading precision of ±10 mV. The reconstruction of Section 4 produced them. Five of these six V_0 values were read off a fitted line and rounded to three decimals. Taken at face value they give

$$\hat{\sigma} = \sqrt{\frac{\sum \hat{\varepsilon}_i^2}{n-2}} = \sqrt{\frac{1.03 \times 10^{-7}}{3}} = 0.18 \text{ mV}, \qquad \text{SE}(\hat{\beta}) = \frac{\hat{\sigma}}{\sqrt{S_{\nu\nu}}} = 6.1 \times 10^{-20} \text{ V·s},$$

a relative precision of 0.0015%. That is three hundred times finer than Millikan claimed.

A plug-in from the instrument. I take σ from the instrument instead of from the residuals. Millikan's balance points were reproducible to about $\sigma = 10\,\mathrm{mV}$. Substituting that value,

$$\mathrm{SE}(\hat{\beta}) = \frac{\sigma}{\sqrt{S_{\nu\nu}}} = \frac{0.010 \text{ V}}{\sqrt{9.277} \times 10^{14} \text{ Hz}} = 3.3 \times 10^{-17} \text{ V·s}, \tag{11}$$

a relative uncertainty of 0.8%, and 0.5% for the six-line design (Section 6.5). Millikan claimed "not more than 0.5%" from graphical fitting. His nine two-point sessions give 0.7% (Section 8). I use $\sigma = 10\,\mathrm{mV}$ throughout.

5.3 Results: Sodium (Six Points, Including 2535 Å)

Including the 2535 Å point ($\nu = 11.827 \times 10^{14}\,\mathrm{Hz}$, $V_0 = 3.030\,\mathrm{V}$—Millikan's explicitly stated true stopping potential) widens the design by a further 59 nm but moves the slope appreciably:

$$\begin{aligned} \bar{\nu} &= 8.240 \times 10^{14} \text{ Hz}, \quad \bar{V}_0 = 1.582 \text{ V}, \\ S_{\nu\nu} &= 24.72 \times (10^{14})^2, \quad S_{\nu V} = 10.06 \times 10^{14} \text{ V·Hz}, \\ \hat{\beta} &= \frac{10.06}{24.72} = 4.070 \times 10^{-15} \text{ V·s}, \end{aligned}$$

1.4% below the five-point estimate, with a residual standard deviation of 11 mV—two orders of magnitude larger than in the five-point fit, and now in line with Millikan's stated reading precision.

The single genuine datum in the table drives the movement. The 2535 Å point lies 0.04 V below the five-point line (Section 4), and its extreme frequency gives it the highest leverage. Millikan attributed the deficit to scattered short-wavelength light in the far UV and excluded the line from his primary determination. The observation that contributes most to $S_{\nu\nu}$ also contributes most to any systematic error in the far UV. Widening the design and protecting it from stray light pull against each other. The 11 mV residual standard deviation here, against 0.18 mV for the five-point fit, confirms that the five-point residuals were rounding artefacts.

5.4 Results: Lithium (Five Points)

Repeating for lithium using lines 4339–2535 Å ($n = 5$, $S_{\nu\nu} = 15.65 \times (10^{14})^2$, $S_{\nu V} = 6.457 \times 10^{14}$):

$$\hat{\beta}_{\text{Li}} = 4.127 \times 10^{-15}\ \text{V·s}, \qquad \hat{\alpha}_{\text{Li}} = -2.352\ \text{V},$$

which differs from the sodium slope by 0.04%.

The agreement is not an independent confirmation of universality. Four of the five lithium entries are the sodium entries shifted by a constant (Section 4), and a constant shift leaves the slope unchanged. Only the 2535 Å line carries information absent from the sodium column. The comparison establishes something weaker. Millikan reported the two metals' stopping-potential curves as parallel at his stated precision.

5.5 Deriving Planck's Constant

With the OLS slope $\hat{\beta} = 4.128 \times 10^{-15}$ V·s (five-point sodium regression) and the elementary charge e from Millikan's oil-drop experiment:

Value of e used	Implied $h = \hat{\beta} \cdot e$	Error
Millikan (1916): 1.592×10^{-19} C	6.573×10^{-34} J·s	−0.81%
Modern: $e = 1.602177 \times 10^{-19}$ C	6.616×10^{-34} J·s	−0.17%
Modern exact: $h = 6.626070 \times 10^{-34}$ J·s	—	0

The error decomposes cleanly. Paired with the modern e, the slope alone accounts for −0.17%; Millikan's own e, too small by 0.64%, contributes the remainder. His h was therefore in error predominantly through the oil-drop constant rather than through the photoelectric slope. His 1916 apparatus was built to measure the slope. Both errors lie inside the 0.8% standard error of equation (11).

5.6 The Slope Planck's Constant Predicted

Einstein derived the slope h/e from Planck's radiation law. Planck fitted h to the blackbody spectrum in 1901 and obtained 6.55×10^{-27} erg·s.[2] Millikan measured $e = 1.592 \times 10^{-19}$ C in his oil-drop experiment. Those two numbers predict a slope

$$\left(\frac{h}{e}\right)_{\text{predicted}} = \frac{6.55 \times 10^{-34}}{1.592 \times 10^{-19}} = 4.114 \times 10^{-15}\ \text{V·s}. \tag{12}$$

The five-point sodium regression gives 4.128×10^{-15} V·s. The measured slope exceeds the predicted slope by 0.35%, inside the 0.8% standard error of (11).

No photoelectric measurement entered the prediction. Planck fitted h to the blackbody spectrum in 1901. Einstein derived the slope in 1905. Millikan measured it in 1916. The regression had no free parameter with which to meet the prediction: Einstein fixed β in advance and left only α free. This comparison, and not the comparison with the modern h in Section 9, is the out-of-sample test of equation (1).

Both 1916-era constants were too small. Planck's h was low by 1.15% and Millikan's e by 0.64%. The errors partly cancel in the ratio. The predicted slope was low by 0.52% and the measured slope by 0.17%. The agreement in (12) therefore rests in part on two errors of the same sign, and a reader who wants the prediction tested against modern constants should compare 4.128 with 4.136×10^{-15} V·s.

6 Maximum Likelihood Estimation of Einstein's Parameters

6.1 The Parametric Model

Einstein's photoelectric equation defines a three-parameter statistical model. The free parameters are:

- $\beta = h/e$ — the universal slope (V·s), encoding Planck's constant;
- $\alpha = -W/e$ — a metal-specific intercept (V), equal to minus the work function in volts under the correction convention of (6);

[2]Planck's 1901 value and the 1913 value in Table 6 both come from secondary sources and should be checked against the originals.

- σ^2 — the variance of the measurement noise ε_i.

The full parametric model for a single metal over n spectral lines is:

$$V_{0i} = \alpha + \beta\,\nu_i + \varepsilon_i, \quad \varepsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0,\sigma^2), \quad i = 1,\ldots,n. \tag{13}$$

The Gaussian assumption is natural: measurement noise in the electrometer readings arises from many small independent sources (thermal Johnson noise, vibration, residual-gas collisions), for which a central-limit argument is compelling.

6.2 The Log-Likelihood

The probability density of a single observation (V_{0i}, ν_i) under model (13) is

$$p(V_{0i} \mid \nu_i;\, \alpha, \beta, \sigma) = \frac{1}{\sqrt{2\pi}\,\sigma}\exp\!\left(-\frac{(V_{0i}-\alpha-\beta\nu_i)^2}{2\sigma^2}\right).$$

For n independent observations the log-likelihood is

$$\ell(\alpha,\beta,\sigma) = -\frac{n}{2}\ln(2\pi) - n\ln\sigma - \frac{1}{2\sigma^2}\underbrace{\sum_{i=1}^{n}(V_{0i}-\alpha-\beta\nu_i)^2}_{\mathrm{RSS}(\alpha,\beta)}. \tag{14}$$

6.3 MLE Equals OLS Under Gaussian Errors

For fixed σ, maximising (14) over (α, β) requires minimising $\mathrm{RSS}(\alpha, \beta)$ — exactly the OLS objective. Therefore:

$$\boxed{(\hat\alpha_{\text{MLE}},\, \hat\beta_{\text{MLE}}) = (\hat\alpha_{\text{OLS}},\, \hat\beta_{\text{OLS}}).} \tag{15}$$

The equivalence holds for any σ.

6.4 MLE for the Noise Variance

Substituting the maximising $(\hat\alpha, \hat\beta)$ into (14) and optimising over σ gives:

$$\hat\sigma^2_{\text{MLE}} = \frac{\text{RSS}}{n}. \tag{16}$$

The OLS estimator uses the degrees-of-freedom correction:

$$s^2 = \frac{\mathrm{RSS}}{n-2}, \tag{17}$$

which is unbiased ($\mathbb{E}[s^2] = \sigma^2$). The MLE $\hat{\sigma}^2_{\mathrm{MLE}}$ is biased downward by the factor $(n-2)/n$; for $n = 5$ the correction is $3/5$.

For the five-point sodium regression, $\mathrm{RSS} = 1.03 \times 10^{-7}\,\mathrm{V}^2$ (from the residuals in Section 5). Hence:

$$\hat{\sigma}_{\mathrm{MLE}} = \sqrt{1.03 \times 10^{-7}/5} = 0.14 \times 10^{-3}\ \mathrm{V} = 0.14\,\mathrm{mV},$$
$$s_{\mathrm{OLS}} = \sqrt{1.03 \times 10^{-7}/3} = 0.18 \times 10^{-3}\ \mathrm{V} = 0.18\,\mathrm{mV}.$$

Both lie nearly two orders of magnitude below Millikan's stated experimental uncertainty of $\pm 10\,\mathrm{mV}$. Sections 4 and 5 give the reason: the residuals are rounding error in the tabulated stopping potentials. I therefore use the instrumental value $\sigma = 10\,\mathrm{mV}$ as the plug-in, not s. The six-line fit, which includes Millikan's one directly quoted stopping potential, gives $s = 11\,\mathrm{mV}$.

6.5 Fisher Information and the Cramér-Rao Bound

The Fisher information matrix bounds the precision attainable by any unbiased estimator of (α, β). For model (13) with σ known, the Fisher information is

$$\mathcal{I}(\alpha, \beta) = \frac{1}{\sigma^2} X^\top X = \frac{1}{\sigma^2} \begin{pmatrix} n & \sum_i \nu_i \\ \sum_i \nu_i & \sum_i \nu_i^2 \end{pmatrix}, \tag{18}$$

where the row $(1,\, \nu_i)$ of the $n \times 2$ design matrix X maps (α, β) into the conditional mean of V_{0i}. The Cramér-Rao bound on the variance of any unbiased estimator of β is

$$\mathrm{Var}(\tilde{\beta}) \geq \left[\mathcal{I}^{-1}(\alpha, \beta)\right]_{\beta\beta} = \frac{\sigma^2}{S_{\nu\nu}}, \tag{19}$$

where $S_{\nu\nu} = \sum_i (\nu_i - \bar{\nu})^2$. The OLS/MLE estimator $\hat{\beta}$ achieves this bound: $\mathrm{Var}(\hat{\beta}) = \sigma^2/S_{\nu\nu}$ (by the Gauss-Markov theorem, OLS is the Best Linear Unbiased Estimator).

For the five-point sodium regression, with $\sigma = 10\,\mathrm{mV}$:

$$\mathrm{SE_{CR}}(\hat{\beta}) = \frac{\sigma}{\sqrt{S_{\nu\nu}}} = \frac{1.0 \times 10^{-2}\ \mathrm{V}}{\sqrt{9.277 \times 10^{14}}\ \mathrm{Hz}} = 3.3 \times 10^{-17}\ \mathrm{V{\cdot}s}, \tag{20}$$

corresponding to a relative precision of 0.79%, which OLS attains exactly. The bound depends on the design and not on the readings. $S_{\nu\nu}$ depends only on the spectral lines Millikan chose. The reconstruction of Section 4 corrupts the residuals and leaves $S_{\nu\nu}$ untouched.

Design implications. By (19), $\mathrm{SE}(\hat{\beta}) \propto 1/\sqrt{S_{\nu\nu}}$. Spreading the frequencies further apart shrinks the standard error whatever the value of n. This justifies Millikan's choice to span nearly a factor of two in ν (5461 Å to 2535 Å), rather than the narrow UV ranges, 60–70 nm wide, available to Hughes and to Richardson & Compton.

Table 4 quantifies the improvement. Restrict Millikan's sodium lines to 4047–3126 Å. That 92 nm window is the narrowest his line set permits and the closest analogue to the ∼60–70 nm bands of Hughes and of Richardson & Compton. It gives $S_{\nu\nu} = 2.44 \times (10^{14})^2\,\mathrm{Hz}^2$, against 9.28 for his five-point design and 24.72 for the full six. Holding σ fixed, the standard error on h/e falls by a factor of $\sqrt{9.28/2.44} = 1.9$ and then $\sqrt{24.72/2.44} = 3.2$. Widening the frequency range improved the design. Collecting more readings would not have.

Table 4: Effect of frequency range on the Cramér-Rao bound for $\hat{\beta} = h/e$, using Millikan's sodium lines and a common instrumental $\sigma = 10\,\mathrm{mV}$. "Narrow" uses the three lines spanning the window closest to the ranges available to Hughes (1912) and Richardson & Compton (1912). Because σ is held fixed, the last two columns isolate the contribution of the experimental design alone.

Design	n	Span (nm)	$S_{\nu\nu}$ ($10^{28}\,\mathrm{Hz}^2$)	$\mathrm{SE}(\hat{\beta})$ ($10^{-17}\,\mathrm{V{\cdot}s}$)	Rel. SE
Narrow (4047–3126 Å)	3	92	2.44	6.4	1.55%
Millikan 5-pt (5461–3126 Å)	5	234	9.28	3.3	0.79%
Millikan full (5461–2535 Å)	6	293	24.72	2.0	0.49%

6.6 Pooled Estimation Across Metals

A more efficient estimator exploits the cross-metal restriction: $\beta = h/e$ is universal. The pooled model is

$$V_{0im} = \alpha_m + \beta\, \nu_i + \varepsilon_{im}, \quad m \in \{\mathrm{Na}, \mathrm{Li}\}, \tag{21}$$

where the intercept α_m is specific to metal m and the slope β is common to Na and Li. The pooled OLS slope estimator is

$$\hat{\beta}_{\mathrm{pool}} = \frac{S_{\nu V}^{\mathrm{Na}} + S_{\nu V}^{\mathrm{Li}}}{S_{\nu\nu}^{\mathrm{Na}} + S_{\nu\nu}^{\mathrm{Li}}}. \tag{22}$$

From Section 5, $\hat{\beta}_{\mathrm{Na}} = 4.128$ and $\hat{\beta}_{\mathrm{Li}} = 4.127$ (both $\times 10^{-15}$ V·s). Pooling by (22) gives

$$\hat{\beta}_{\mathrm{pool}} = \frac{3.830 + 6.457}{9.277 + 15.647} = 4.127 \times 10^{-15}\ \mathrm{V{\cdot}s},$$

with $S_{\nu\nu}$ raised from 9.28 to $24.92 \times (10^{14})^2$ and hence $\mathrm{SE}(\hat{\beta}_{\mathrm{pool}}) = \sigma/\sqrt{24.92} \times 10^{-14} = 2.0 \times 10^{-17}$ V·s, a relative precision of 0.49%. Pooling two metals buys what adding the sixth sodium line buys. Both enlarge $S_{\nu\nu}$.

The Wald statistic for testing the slope-equality restriction $H_0\colon \beta_{\mathrm{Na}} = \beta_{\mathrm{Li}}$ is

$$z = \frac{\hat{\beta}_{\mathrm{Na}} - \hat{\beta}_{\mathrm{Li}}}{\sqrt{\mathrm{SE}^2(\hat{\beta}_{\mathrm{Na}}) + \mathrm{SE}^2(\hat{\beta}_{\mathrm{Li}})}} \approx \frac{0.002 \times 10^{-15}}{\sqrt{2} \times 3.3 \times 10^{-17}} \approx 0.04,$$

far below the critical value of 1.96. The restriction is not rejected.

The test has little power. With a standard error of 0.8% on each slope it would fail to reject slope differences of up to about 2%, so it excludes only gross violations of universality. The test is also partly circular. Four of the five lithium observations are the sodium observations shifted by a constant (Section 4), which forces $\hat{\beta}_{\mathrm{Li}}$ toward $\hat{\beta}_{\mathrm{Na}}$. The near-zero z reflects that construction. Millikan reported parallel V_0–ν curves for the two metals at his stated precision. Nothing in the pooled data contradicts universality. These ten numbers do not establish it.

7 The Contact-EMF Identification Problem

The contact EMF v_c between the photoemitting metal and the collector is a nuisance parameter. In the raw readings of (4) the intercept is $-W/e - v_c$, which conflates the work function with an apparatus term that was often of comparable size and was not stable across the early experiments.

Pre-Millikan experiments. The early experiments did not measure or control v_c. If v_c drifted between measurements at different frequencies, the apparent stopping potentials would shift inconsistently and bias both slope and intercept. Hughes and Richardson & Compton measured different lines on different days with no guarantee of constant v_c, which Millikan [1916] identified as a contributing cause of their h values being 12–47% below Planck's.

Millikan's solution. Millikan measured all six spectral lines in a single continuous session after a freshly shaved surface had stabilised (typically 1–2 hours after shaving). He verified that v_c remained constant to $\pm 0.01\,\text{V}$ over the session by using the Kelvin double-balance. When v_c is constant across all observations, it enters as a pure intercept shift and does not affect $\hat{\beta}$. The slope $\hat{\beta} = h/e$ is then estimated without bias regardless of the unknown value of v_c.

Separate identification of W. The independent Kelvin measurement $v_c = 2.51\,\text{V}$ is what makes the intercept interpretable at all. Having used it to convert his raw readings into true stopping potentials, Millikan could read the work function directly off the fitted intercept $\hat{\alpha} = -1.813\,\text{V}$ of equation (10), since by (6)

$$W_{\text{Na}}/e = -\hat{\alpha} = 1.81 \text{ eV},$$
$$W_{\text{Na}} = 1.81 \times 1.602 \times 10^{-19} = 2.90 \times 10^{-19} \text{ J}.$$

The corresponding lithium figure is $-\hat{\alpha}_{\text{Li}} = 2.35\,\text{eV}$. The dependent variable already contains v_c. Adding it again double-counts it.

These estimates are the weakest quantities in the analysis. The modern value for clean bulk sodium is $2.36\,\text{eV}$, and the intercept-implied figure is low by about $0.5\,\text{eV}$. Estimation error accounts for part of the gap. The intercept extrapolates to $\nu = 0$, far outside the observed range (5.5–$11.8 \times 10^{14}\,\text{Hz}$), and its standard error is $\sigma\sqrt{1/n + \bar{\nu}^2/S_{\nu\nu}} \approx$

0.08 V at $\sigma = 10\,\mathrm{mV}$. Surface physics accounts for more. Millikan's shaved surfaces sat in 10^{-2} mmHg of residual gas, and adsorbed gas shifts a metal's work function by several tenths of a volt. W varies by some 0.3 eV across the crystal faces of a single metal. Sodium has no one work function for the intercept to be compared against.

The two coefficients are identified on different terms. The slope is a universal constant. Variation within a run identifies it, and any disturbance common to all lines leaves it alone. The intercept is a surface-specific quantity. Extrapolation identifies it, and the conditions Millikan could not control contaminate it. Einstein's equation is better identified in its slope than in its intercept. The slope carries the physics.

8 Nine Two-Point Determinations

The five- and six-point regressions cannot supply a credible standard error from their own residuals (Section 4). Millikan's Table II can. He estimated the slope from nine pairs of spectral lines, always pairing one line with 5461 Å, chosen for its high intensity and its precise balance point. Each row is a separate session with a freshly shaved surface. The nine values are replications. Their dispersion measures what varied from run to run: surface condition, residual gas, drift in v_c, reading error.

The confidence interval is Student-t with eight degrees of freedom ($t_{.975,8} = 2.306$); the normal approximation would give [4.073, 4.189].

The two-point mean 4.131×10^{-15} V·s agrees with the five-point OLS estimate 4.128×10^{-15} V·s to better than 0.1%. The reconstruction of Section 4 has not distorted the slope. The session-to-session standard deviation of 0.087, or 2.1% of the mean, is the scale of the experiment's variability. It exceeds the 0.18 mV residual scatter of the five-point fit by a factor of about a hundred. The standard error of the mean, 0.029×10^{-15} or 0.71%, estimates the precision of Millikan's determination from repeated measurement alone.

That figure sits on the Cramér-Rao bound of Section 6.5, which the design and $\sigma = 10\,\mathrm{mV}$ give as 0.79% for the five-line span. Replication and the information matrix agree to within a tenth of a percent. This agreement warrants the uncertainty attached to h in this paper. The 1–4% spread of individual sessions is what $\sigma \approx 10\,\mathrm{mV}$ applied to a two-point slope produces.

Table 5: Millikan (1916), Table II: nine two-point slope determinations for sodium. Each row is an independent measurement session with a fresh surface shaving. The mean and standard deviation are computed from these nine observations.

Session	λ paired with 5461 Å	$\hat{\beta}$ (10^{-15} V·s)
1	3126 Å	4.11
2	3650 Å	4.14
3	3126 Å	4.10
4	3650 Å	4.12
5	3126 Å	4.24
6	4047 Å	3.98
7	2535 Å	4.04
8	3126 Å	4.24
9	4047 Å	4.21
Mean		**4.131**
Std. dev. ($n = 9$)		0.087
Std. error of mean		0.029
95% CI		[4.06, 4.20]

9 Comparison of Historical Estimates

Table 6 compares the estimates of h from all major pre-1920 photoelectric experiments.

Millikan's photoelectric determinations sit 0.7–0.9% below the modern value. His oil-drop charge, too small by 0.64%, accounts for almost all of that gap. Rerun with the modern e, every one of his slopes lands within 0.22% of the defined h, inside the 0.8% standard error of the design. His photoelectric measurement was more accurate than the number he published.

The pre-Millikan estimates are biased downward by 12–47%. The discrepancy exceeds any plausible sampling error by two orders of magnitude, so it is systematic. Three mechanisms produce it. Stray short-wavelength light raises the apparent V_0 at long-wavelength lines and flattens the slope. A restricted frequency range inflates the standard error, as Table 4 quantifies, and leaves the estimate at the mercy of a single bad line. Contact EMF that drifts between sessions contaminates observations individually instead of shifting a common intercept, and biases the slope in an unpredictable direction. Millikan's engineering addressed all three: fresh surfaces, filtered lines, a wide frequency range, and continuous Kelvin monitoring of v_c. Only the second is a statistical problem.

Table 6: Historical estimates of Planck's constant h from photoelectric experiments. Millikan's attribution of earlier errors is discussed in Millikan [1916, pp. 373–375]. For the Millikan rows the slope $\hat{\beta}$ is converted to h twice: once with the charge Millikan himself had measured, $e = 1.592 \times 10^{-19}$ C, and once with the modern $e = 1.602177 \times 10^{-19}$ C. Errors are relative to the defined value $h = 6.626070 \times 10^{-27}$ erg·s. The two Planck values come from secondary sources and should be checked against the originals.

Authors	Year	Metal(s)	h, Millikan's e	Err.	h, mod. e
Hughes	1912–13	various	3.55–5.85	−12 to −46%	—
Richardson & Compton	1912	8 metals	3.5–5.8	−12 to −47%	—
Millikan (Na, 5-pt)	1916	Na	6.573	−0.81%	6.615
Millikan (Na, 9 two-pt)	1916	Na	6.577	−0.74%	6.619
Millikan (Li, 5-pt)	1916	Li	6.570	−0.85%	6.612
Millikan (Na+Li pooled)	1916	Na+Li	**6.571**	**−0.83**%	**6.613**
Planck (black body)	1901	—	6.55	−1.15%	—
Planck (black body)	1913	—	6.415	−3.19%	—
Modern (defined)	2019	—	6.626070	0	6.626070

Estimation theory cannot repair the other two after the fact.

10 Millikan's Reluctance

Millikan's 1916 paper confirmed Einstein's equation (1) to to better than 1%. Yet Millikan wrote:

> "Despite then the apparently complete success of the Einstein equation, the physical theory of which it was designed to be the symbolic expression is found so untenable that Einstein himself, I believe, no longer holds to it." [Millikan, 1916, p. 388]

The light-quantum hypothesis was, in Millikan's judgment, physically implausible because it conflicted with the wave theory of diffraction and interference. He had hoped his experiment would *refute* the equation, and was surprised when it confirmed it.

Compton scattering [Compton, 1923] supplied, in 1923, the independent evidence for the particulate character of light that the photoelectric regression could not. Millikan came to describe his own result in these terms:

> "This work resulted, contrary to my own expectation, in the first direct experimental proof . . . of the complete validity of the Einstein equation."

The sentence belongs to Millikan's later retrospective writing, not to the Nobel lecture he delivered in May 1924. Holton [1999] and Pais [1982] trace the trajectory of his views.

A macroeconomist likewise can confirm a reduced-form predictive relationship without endorsing the structural model that motivated it.

11 Einstein as Machine Learner

11.1 Overview

The vocabulary of modern machine learning—supervised learning, hypothesis class, loss function, inductive bias, feature engineering, transfer learning—did not exist in 1905. The operations Einstein performed are the ones that statistical learning theory formalises. Sargent [2025] argues that this is the rule rather than the exception, and that the techniques now gathered under artificial intelligence have long histories inside the sciences that used them. This section states the correspondence for the photoelectric case and its limits.

11.2 Supervised Learning and Labeled Data

The photoelectric dataset is a standard supervised regression problem. The *input feature* is the frequency ν of the incident light; the *label* is the stopping potential V_0. The task is to learn a function $f : \nu \mapsto V_0$ that generalises to unobserved frequencies.

Lenard (1902) provided the *qualitative* version: the label was observed (high V_0 at high frequency) but not quantified reliably. Millikan (1916) provided the *quantitative* training set: six labelled pairs (ν_i, V_{0i}) for sodium and five for lithium, with precision sufficient to estimate the slope h/e to under 1%.

11.3 The Hypothesis Class and Inductive Bias

Einstein's hypothesis class is:

$$\mathcal{H} = \left\{ f_{\alpha,\beta} : \nu \mapsto \alpha + \beta\nu \;\middle|\; \alpha \in \mathbb{R},\ \beta > 0 \right\}, \tag{23}$$

together with the *cross-metal constraint* that β is the same for all metals. The class is extremely tight. For a single metal it has two free parameters, (α, β). Across M metals the constraint bites: instead of $2M$ free parameters there are $M + 1$—one intercept apiece and a single shared slope—so the class is a family of parallel lines whose common gradient is the one physical unknown.

The restriction to $\mathcal{H}$ was not ad hoc. It followed deductively from the quantum hypothesis $E = h\nu$ and energy conservation for photoemission. The quantum hypothesis *is* the inductive bias: it restricts the hypothesis class from all possible functions of $(\nu, I, \lambda, \ldots)$ to a specific affine form in ν alone. Without this bias five data points are hopelessly insufficient: infinitely many smooth functions pass through five points.

11.4 Feature Engineering: Choosing ν Over Intensity and Wavelength

Learning begins with a choice of input features. Lenard's experiments performed a feature importance analysis. He measured V_0 as a function of both frequency (colour) and intensity I. His result was unambiguous: V_0 varies strongly with ν but not at all with I.

Classical wave theory predicted the opposite: wave amplitude $A \propto \sqrt{I}$ should determine energy delivered to an electron, so V_0 should grow with I. The empirical irrelevance of I was the anomaly that motivated Einstein's model. Lenard's experiment eliminated I from the feature set and selected ν before any parametric fitting was done.

11.5 The Loss Function and Training

Given the hypothesis class (23) and the feature ν, choose (α, β) to minimise the empirical risk under squared-error loss:

$$\hat{R}(\alpha, \beta) = \frac{1}{n} \sum_{i=1}^{n} (V_{0i} - \alpha - \beta\nu_i)^2.$$

This is identical to the MSE loss used in neural-network regression. The minimiser of $\hat{R}$ is the OLS/MLE estimator derived in Sections 5–6: training is the closed-form solution $\hat{\beta} = S_{\nu V}/S_{\nu\nu}$. For Einstein's linear model, the loss surface is a convex paraboloid in (α, β) with a unique global minimum; no stochastic gradient descent is needed.

11.6 Generalisation and Transfer Learning

Generalisation. The fitted line $\hat{V}_0 = \hat{\alpha} + \hat{\beta}\nu$ predicts stopping potentials for frequencies *not in the training set*. The residuals of the five-point fit cannot establish this, since they measure rounding (Section 4). Two other results can. The six-point fit, which includes a line withheld from the five-point design, leaves a residual standard deviation of 11 mV at that line. The nine two-point sessions of Section 8 each fit two lines and agree with the five-point slope to 0.1%. In learning-theoretic terms, the relevant capacity measure for a real-valued class is the pseudo-dimension (the Vapnik-Chervonenkis dimension is defined for classifiers); for affine functions of a single input it equals 2, the number of free parameters. Five training points against a capacity of two leave three residual degrees of freedom—enough both to fit and to assess the fit, which is precisely what a capacity of five or more would forbid.

Transfer learning. A model trained on sodium data successfully predicts lithium stopping potentials up to an intercept shift. The slope $\hat{\beta} = h/e$ is transferred exactly from one metal to another. Training on sodium provides a pre-trained feature extractor, the slope; the lithium task fine-tunes the intercept α_{Li}, the metal-specific work-function parameter. The Wald test of Section 6.6 confirms that the zero-shot transfer of the slope from Na to Li is not rejected by the data ($z = 0.04$), subject to the two cautions recorded there.

11.7 Model Selection: Quantum vs. Classical Wave Theory

At the time of Einstein's paper two models were in competition:

- **Classical energy-transfer model.** The energy delivered to an electron is set by the wave amplitude $A \propto \sqrt{I}$ accumulated over the illumination time. V_0 should therefore rise with intensity, rise with exposure, and depend on frequency only weakly and through no prescribed functional form.

- **Einstein's model.** $V_0 = (h/e)\nu - W/e$: exactly linear in ν, wholly independent of I, with a slope fixed *a priori* at h/e and common to every metal.

On these data the second model wins by any criterion. The data bear on three claims, and the evidence for them differs.

Lenard settled the irrelevance of intensity before anyone ran a regression, and that alone refutes the classical energy-transfer picture. The regression establishes linearity in ν. The numerical value of the slope is the sharpest test. Planck's blackbody constant fixed h/e in advance, Einstein's prediction had no free parameter to adjust, and the measured slope came within 0.35% of it (Section 5.6). A model-selection criterion that scores in-sample fit credits a correct prediction and a lucky post-hoc fit alike, and misses this last point.

The data do not select the light-quantum mechanism. Equation (1) follows from energy conservation together with quantised *atomic* energy levels, and a semiclassical model in which the radiation field remains a classical wave delivers it too (Appendix A.4). The regression identifies the best-fitting member of a hypothesis class and rules out a rival class. It cannot discriminate between two physical stories that imply the same class. Photon-number statistics discriminated between them sixty years later. Model selection has this limitation generally, and Millikan's execution of it is not at fault.

11.8 What a Naive Machine Learning System Would Miss

A purely data-driven system, given only the table of (ν_i, V_{0i}) pairs, would learn the following.

1. A linear regression package would correctly recover $\hat{\beta} = 4.128 \times 10^{-15}$ V·s.
2. It would *not* know that $\hat{\beta}$ identifies a fundamental constant of nature. The number 4.128×10^{-15} is physically meaningless without the quantum hypothesis that gives it the interpretation h/e.
3. It would *not* impose the cross-metal universality restriction $\beta_{\mathrm{Na}} = \beta_{\mathrm{Li}}$ from first principles. The universality is a prediction of the quantum hypothesis; from a purely data-driven perspective, it is just a coincidence to be discovered, not a constraint to be imposed.
4. It would have no way to decompose the intercept into the physical work function

$-W/e$ and the apparatus nuisance v_c, since the two are not separately identified without the external Kelvin measurement (Section 7). Identification failures of this kind are invisible to a fitting procedure: the regression runs, reports an intercept, and says nothing about what the intercept means.

Section 12 pursues the comparison to a system with far greater capacity than a linear regression package, and reaches the same conclusion by a different route.

Einstein supplied the theoretical framework that gives the regression coefficients their physical meaning. Specifying the hypothesis class required more intellectual effort than fitting the line.

11.9 The Deductive–Inductive Loop

Modern discussions of "AI for science" often describe a purely inductive paradigm: given enough data, learn the law. The photoelectric episode follows the hypothetico-deductive loop that has driven physical science since Newton.

Table 7: The hypothetico-deductive loop in the photoelectric episode, translated into the language of modern machine learning.

Step	Scientific name	ML name
Propose $E = h\nu$	Theoretical hypothesis	Model architecture
Derive $V_0 = (h/e)\nu - W/e$	Deduction	Hypothesis class
Choose ν not I	Feature selection	Feature engineering
Measure (ν_i, V_{0i})	Experiment	Training data
Min. $\sum(V_{0i} - \hat{V}_{0i})^2$	Least squares	MSE training
$\lvert\hat{\beta} - h/e\rvert < 1\%$	Confirmation	Test accuracy
Na and Li: same $\hat{\beta}$	Universality	Zero-shot transfer

Five training points estimate h to 0.1% because the physics prescribes the functional form, the number of free parameters (α, β), and the units of the slope. A purely inductive system that must discover the functional form from data faces a far larger sample-complexity problem.

12 A Deep Neural Network on the Photoelectric Data

12.1 Two Hypothesis Classes Compared

Einstein's model and a deep neural network (DNN) represent opposite extremes of the bias-variance spectrum for the photoelectric regression. Einstein's hypothesis class is equation (23): two free parameters per metal, one shared slope across metals, pseudo-dimension 2. The Gauss–Markov theorem and the Cramér–Rao bound together guarantee that OLS achieves minimum variance among all unbiased estimators; five training points are more than sufficient.

A DNN with L hidden layers and N neurons per layer is a universal function approximator: for any continuous target function on a compact interval, there exists a network that approximates it to arbitrary precision [Hornik et al., 1989]. For a one-dimensional input (ν) and one-dimensional output (V_0), even a modest architecture—three hidden layers with 16 ReLU neurons each—has approximately

$$p = (1 \cdot 16 + 16) + (16 \cdot 16 + 16) + (16 \cdot 16 + 16) + (16 \cdot 1 + 1) = 593$$

free parameters. For ReLU networks the pseudo-dimension grows like $p \log_2 p$, here $593 \log_2 593 \approx 5{,}500$ [Vapnik, 1998].

The literature offers no single conversion from a capacity figure to a required sample size. The uniform-convergence bound $n = O(d/\epsilon^2)$, with d the pseudo-dimension, gives $n \sim 10^7$ for $\epsilon = 1\%$. The bound is vacuous at that size. The practitioner's rule $n \gtrsim 10p$ gives $n \gtrsim 6{,}000$. I use the rule throughout as the "sufficient n" for the DNN. The rule is a heuristic. The formal bound is four orders of magnitude worse.

Capacity bounds of this kind are loose. Overparameterised networks generalise in regimes where uniform-convergence theory predicts they cannot [Belkin et al., 2019], and the argument below does not rest on a claim that a DNN provably fails. The difficulty at $n = 5$ concerns identifiability and not generalisation error. Section 12.4 makes it precise. The comparison with Einstein's two-parameter model turns on degrees of freedom, not on a VC bound.

12.2 Epoch I—Lenard's Qualitative Findings (1902)

Lenard established two qualitative facts: stopping potential rises with frequency and is independent of intensity. Lenard recorded no (ν_i, V_{0i}) pairs. A DNN requires labelled numerical examples; with none, it cannot begin training. Einstein required zero labelled examples: the quantum hypothesis plus energy conservation yielded equation (1) as a deduction. Einstein's "model fitting" was a one-paragraph derivation. The network's fitting cannot begin.

12.3 Epoch II—Hughes and Richardson & Compton (1907–1914)

The intermediate experiments produced roughly 25–40 numerical (ν_i, V_{0i}) pairs, clustered in a narrow 60 nm UV range with uncontrolled contact EMFs and stray scattered light. Three problems foreclose reliable DNN learning.

Underdetermination. With $n \approx 30$ and $p = 593$, the DNN is massively overparameterised. Gradient descent finds a minimum-norm interpolant that passes through all 30 training points exactly (zero training loss), but is effectively unconstrained between them. Applying ℓ_2 weight decay suppresses oscillations, but the optimal regularisation strength cannot be selected without a held-out validation set; with $n = 30$ this is statistically isolated.

Restricted frequency range. A linear, a quadratic, and a square-root form fit almost equally well over a 60–70 nm UV band ($\Delta\nu \approx 1.5 \times 10^{14}$ Hz). The DNN has no mechanism to infer which of them is the global functional form.

Systematic label contamination. Uncontrolled contact EMFs imply that the effective label for observation i is $V_{0i} + \delta_i$, where δ_i is unknown and observation-specific. The DNN receives contaminated targets and, lacking any physical model of the contaminant, cannot separate the linear physical trend from the systematic noise. The learned function absorbs both, producing a biased and uninterpretable output.

12.4 Epoch III—Millikan's Data (1916)

Millikan's sodium experiment was the cleanest of the era. It yielded five usable (ν_i, V_{0i}) pairs for the primary determination.

The interpolation regime. With $n = 5$ and any DNN with $p > 5$ (which includes every useful architecture), the network operates in the interpolation regime: it memorises all five training points exactly, producing zero training loss [Belkin et al., 2019]. The resulting function is unconstrained everywhere between the training points. Even the best-case minimum-norm interpolant—"well-behaved" in the double-descent sense—provides no basis for slope inference: there are no residual degrees of freedom with which to estimate the noise variance or a standard error on the slope.

Loss of inferential machinery. OLS with $n = 5$ and $p = 2$ free parameters uses $n - 2 = 3$ residual degrees of freedom to estimate $\hat{\sigma}^2$ and hence $\mathrm{SE}(\hat{\beta})$. Any DNN with more than three parameters exhausts those degrees of freedom entirely. The DNN cannot report a confidence interval for h/e, cannot test whether the slope differs from zero, and cannot quantify the 0.1% relative precision that OLS achieves at the Cramér–Rao bound.

No basis for universality. The universality of the slope $\beta = h/e$ across metals is a theoretical prediction of the quantum hypothesis, not an empirical feature discoverable from sodium data alone. A DNN trained on sodium would need to be retrained from scratch on lithium. The only mechanism by which the slope could transfer automatically from Na to Li is an explicit architectural constraint that encodes the universality—which is equivalent to importing the physics into the model.

12.5 The Modern Era: Large Datasets

By the mid-twentieth century, photoelectric measurements spanned decades of metals, surface preparations, and photon energies. Pooling even a modest subset—say $n = 5{,}000$ clean observations over two octaves of frequency for a single metal—brings a DNN into the classical regime ($n \gg p$), if only barely on the reckoning above.

With smooth activations and appropriate regularisation, the DNN would converge, as $n \to \infty$, to the minimum-population-risk function in its hypothesis class. Because the true function is exactly linear, the minimum-risk function in any sufficiently rich class *is* the linear function. The DNN would eventually recover the line. It would still not identify the constant, impose the cross-metal restriction, decompose the intercept, or match Einstein's sample size:

- Identify the learned slope $\hat{w} \approx 4.128 \times 10^{-15}\,\mathrm{V{\cdot}s}$ as the ratio h/e of two independently measured constants. The slope appears as a numerical weight, not as a ratio of fundamental quantities.
- Impose the cross-metal constraint $\beta_{\mathrm{Na}} = \beta_{\mathrm{Li}} = \cdots$ from first principles. Universality would appear as an empirical pattern to be confirmed, not a structural prior to be imposed.
- Decompose the intercept into a physical work function W/e and an apparatus nuisance v_c without independent measurements of v_c.
- Reduce the required sample from $O(10^3)$ examples to five. A correctly specified two-parameter linear model needs five points; no architectural innovation can replicate this compression without importing the physics.

Modern symbolic regression systems such as AI Feynman [Udrescu and Tegmark, 2020] can, in principle, rediscover the linear form and even match $\hat{w}$ to h/e if given units and physical constants as hardcoded inputs. But that hardcoding supplies prior physical knowledge. The (ν, V_0) data do not contain it.

12.6 Summary: Sample Complexity and the Value of Theory

Table 8 quantifies the contrast.

DNNs are good learners for high-dimensional problems in which no parsimonious parametric family is known: image recognition, protein structure prediction, language modelling. The photoelectric problem sits at the opposite extreme: a one-dimensional input, a one-dimensional output, and five to thirty training examples.

Einstein's proposal $E = h\nu$ reduced the hypothesis space from the infinite-dimensional function class of a DNN to a two-dimensional manifold of parallel lines, and with it the required number of training examples by three orders of magnitude. He did so before a single labelled example was available.

Table 8: Sample-complexity comparison between Einstein's two-parameter linear model (pseudo-dimension = 2, derived from the quantum hypothesis) and a three-layer DNN with 16 neurons per layer ($p = 593$ parameters, pseudo-dimension $\approx$ 5,500). "Sufficient n" uses the practitioner's rule $n \gtrsim 10p \approx$ 6,000; the formal uniform-convergence requirement $O(d/\epsilon^2)$ is some four orders of magnitude larger and is not used here.

Epoch	n available	Einstein model	DNN (VC$\approx$5700)	DNN outcome
Lenard 1902	≈ 0	Sufficient†	Insufficient	Cannot train (no labels)
Hughes 1912	~ 30	Sufficient	Insufficient ($\times$200)	Memorises; no generalisation
Millikan 1916	5	Sufficient	Insufficient ($\times$1200)	Interpolates; no inference
Modern era	~ 5000	Sufficient	Short by $\times$1.2	Recovers linear shape; cannot identify h/e

† The functional form is derived from theory; only two free parameters remain to be estimated from data.

13 Conclusions

Einstein's photoelectric equation is a single-regressor linear model with one slope (h/e) and one intercept ($-W/e$, once the contact EMF has been measured and removed). The regression is trivial given clean data. Obtaining clean data was the central challenge. Fresh metal surfaces free of oxide eliminate work-function noise. High vacuum eliminates atmospheric contamination. Spectrally pure light eliminates bias from stray UV. A wide frequency range reduces the standard error of the slope. Control of v_c identifies the intercept.

Millikan's decade-long experimental program was a campaign to raise data quality until the OLS standard errors were small enough to distinguish a specific value of the slope ($h/e = 4.136 \times 10^{-15}$ V·s) from the alternatives implied by earlier, discrepant theories.

The final five-point sodium regression gives

$$\hat{\beta} = 4.128 \times 10^{-15} \text{ V·s} \pm 0.8\% \quad (h = 6.62 \times 10^{-34} \text{ J·s at the modern } e,\ 0.17\% \text{ low}),$$
$$\hat{\alpha} = -1.813 \text{ V} \quad (W_{\text{Na}}/e = 1.81 \text{ eV}).$$

The quoted standard error is the Cramér-Rao bound at Millikan's instrumental precision of 10 mV, not the residual-based figure, which the reconstruction of the tabulated potentials renders meaningless (Section 4). It agrees with the 0.7% standard error of his nine independent two-point sessions, and with his own claim of "not more than 0.5%" for the full six-line design.

Five observations sufficed to measure a fundamental constant to under one percent. The physics had already fixed the functional form, the number of free parameters, and the units of the slope. A learner obliged to discover those from data would need thousands of observations and would still not know what it had found. Theory compresses the sample size that data must supply.

Millikan's fit selected Einstein's functional form over its classical rival. It could not select the light-quantum mechanism. The same equation follows from a semiclassical model in which the field is never quantised (Appendix A.4). Photon-antibunching experiments settled that question sixty-one years later (Appendix A.5). No one would have performed them without the quantum theory of optical coherence to say which observable mattered. Theory sets the hypothesis class. Theory then tells the experimenter which measurement discriminates within it.

A Einstein's Equation in Modern Physics

A.1 The Equation Remains Exact

Einstein's photoelectric equation (1) has not been superseded. The equation states energy conservation for the fastest photoelectron, the electron that starts at the Fermi level and reaches the detector without inelastic loss. It survives intact in the quantum-electrodynamical treatment, where single-photon photoemission is a first-order process whose energy balance is exactly (1).

The word applies to the energy balance and not to the measured spectrum. A real metal broadens the photoemission edge over $\sim k_B T$, so the stopping potential locates an edge and not a sharp cutoff. W is a property of the crystal face and of the adsorbate layer, not of the element. Final-state relaxation shifts core-level energies by amounts that matter at spectroscopic precision. The work function W is the energy required to move an electron from the Fermi level to the vacuum level, and density-functional theory computes it to a few tenths of an eV for clean surfaces. Millikan's regression tests the energy balance, which is exact; the residual scatter he saw came from the surface physics, which is not.

The equation is exploited continuously in surface science. Ultraviolet Photoelectron Spectroscopy (UPS) and Kelvin probe instruments apply equation (1) to measure work functions of semiconductor films, organic photovoltaic layers, and heterogeneous catalysts, routinely resolving differences of a few meV.

A.2 X-ray Photoelectron Spectroscopy

When the incident photon energy $h\nu$ is drawn from the X-ray range (100 eV–10 keV), the emitted electron's kinetic energy eV_0 identifies the *binding energy* E_B of the core electron level from which it was ejected [Siegbahn et al., 1967]:

$$E_B = h\nu - eV_0 - \phi_{\text{spec}}, \tag{24}$$

where ϕ_{spec} is the spectrometer work function, a calibration constant. Equation (24) is equation (1) rearranged: the unknown is now the binding energy rather than h/e. Because each element has a unique set of core-level binding energies, X-ray Photoelectron Spectroscopy (XPS/ESCA) provides quantitative, element-specific chemical analysis of surfaces. Kai Siegbahn received the 1981 Nobel Prize in Physics for this technique, which is now one of the most widely used methods in materials science, semiconductor device fabrication, and surface catalysis.

In statistical terms, XPS is a multivariate extension of Millikan's regression: dozens of binding energies are estimated simultaneously by inverting Einstein's linear equation at multiple photon energies.

A.3 Multi-Photon Photoelectric Effect

Einstein's equation assumes each emitted electron absorbs exactly one photon. At the peak intensities of pulsed lasers ($I \gtrsim 10^{10}\,\text{W/cm}^2$), an electron can sequentially absorb $n > 1$ photons before escaping the surface. The n-photon generalisation of equation (1) is

$$eV_0 = n\,h\nu - W, \qquad n = 1, 2, 3, \ldots \tag{25}$$

Above-threshold ionisation (ATI)—photoelectron emission at kinetic energies exceeding the single-photon threshold by integer multiples of $h\nu$—was first observed by Agostini et al. [1979]. The resulting photoelectron spectrum displays a comb of peaks separated by $h\nu$, each satisfying equation (25) for the corresponding n. ATI extends hypothesis class (23) to allow non-unit photon multiplicity. The n-photon branch has slope nh/e, not h/e: the class becomes a family of lines whose gradients are integer multiples of a single universal constant, with the metal still entering only through the common intercept. In regression terms this is a mixture model over a discrete latent index n. The gradients are quantised rather than free, and a single parameter governs every branch.

A.4 Does the Photoelectric Effect Prove the Quantisation of Light?

Lamb and Scully [1969] showed that the photoelectric stopping-potential equation can be derived from a *semiclassical model*: the electromagnetic field is treated as a classical wave while only the atom (electron) is quantised. In this model a classical wave drives transitions between quantised atomic energy levels, and the ejected-electron kinetic energy still satisfies equation (1). The reason is that equation (1) follows from energy conservation and the quantised *atomic* level structure alone; it does not require the radiation field itself to be quantised.

Millikan's regression confirms the functional form $V_0 \propto \nu$ and measures h/e to better than 1%. It does not distinguish the quantised-field hypothesis from semiclassical alternatives. The slope h/e gives Planck's constant whether or not the field is composed of quanta.

A.5 Photon Antibunching: Definitive Evidence for Field Quantisation

The genuine experimental proof that the radiation field must be quantised came not from the photoelectric effect but from photon-counting experiments in quantum optics. The relevant observable is the statistical regularity—or irregularity—with which successive photons are emitted.

Hanbury Brown and Twiss [1956] observed *photon bunching* in thermal (chaotic) light: photon pairs from a thermal source arrive at two detectors with a coincidence rate exceeding a Poisson benchmark, reflecting the Bose-Einstein statistics of thermal photons. Glauber [1963] developed the quantum theory of optical coherence, introducing coherent states of the radiation field (the states most closely resembling classical waves) and deriving the photon-number statistics that separate classical, coherent, and thermal light. Glauber received the 2005 Nobel Prize in Physics for this framework.

The decisive evidence for field quantisation is photon *antibunching*: the observation that successive photons from a single quantum emitter are *less* likely to arrive together than Poisson statistics predict. Kimble et al. [1977] first observed sub-Poissonian photon-number statistics in the resonance fluorescence of individual sodium atoms. Antibunching is impossible for any classical stochastic field: such a field, however modulated or noisy, yields a photocount distribution at least as broad as Poisson, so sub-Poissonian statistics have no classical representation. The observation therefore refutes the *neoclassical* radiation theories that sought to dispense with field quantisation altogether, and establishes that the radiation field is quantised.

The result does not refute Lamb and Scully. He argued that the photoelectric effect by itself does not compel field quantisation, and antibunching leaves that claim standing. The measurement of Kimble et al. [1977] required correction for fluctuations in the number of atoms in the observation volume, which by themselves produce super-Poissonian counts. Cleaner demonstrations with single trapped ions followed in the 1980s.

Einstein's 1905 paper introduced the photon as a *heuristic*—his own word in the title—and Millikan confirmed its equation to better than 1% in 1916. Definitive experimental evidence that the radiation field is itself quantised arrived only in 1977, by a route—photon-number statistics in resonance fluorescence—that Einstein's equation neither anticipated nor directly implied. The regression confirmed the equation; quantum optics experiments seventy years later confirmed the physical interpretation.

A.6 Implications for the Machine Learning Reinterpretation

The modern results above bear on Section 11.

Depth of the hypothesis class. Einstein chose hypothesis class (23) by a heuristic argument whose full justification—QED, Fermi's golden rule, the Fermi-Dirac distribution of electrons in a metal—emerged over the following three decades. The class proved correct for reasons deeper than those available in 1905, and it extends over a five-order-of-magnitude range of photon energies, from visible UV to hard X-rays, without modification.

What the regression tests. Millikan's regression tests the functional form $V_0 \propto \nu$ and estimates the slope h/e. Both are consistent with the quantised-field and with the semiclassical model of radiation. The regression identifies the best-fitting model within a hypothesis class; selecting between the two classes requires a different observation, photon-number variance, and a different experimental design.

Theory specificity in experimental design. The antibunching experiments of Kimble et al. [1977] were designed specifically to test the quantised-field hypothesis; the relevant observable (photon-number variance) follows directly from Glauber's coherence theory. Without the quantised-field hypothesis and its prediction of sub-Poissonian statistics, there was no reason to measure photon-number correlations at all. Theory specifies the hypothesis class and, from it, the decisive test. Each theoretical refinement between Millikan (1916) and Kimble et al. (1977) generated a new and more discriminating experiment. Theory forged the links in that chain of hypothetico-deductive loops (Table 7).

References

Pierre Agostini, François Fabre, Gilles Mainfray, Gérard Petite, and Nassir K. Rahman. Free-free transitions following six-photon ionization of xenon atoms. *Physical Review Letters*, 42(17):1127–1130, 1979. doi: 10.1103/PhysRevLett.42.1127. First observation of above-threshold ionisation (ATI): photoelectrons with kinetic energies $eV_0 = nh\nu - W$ for integer $n > 1$; extends Einstein's single-photon equation to multi-photon absorption at high laser intensities.

Mikhail Belkin, Daniel Hsu, Siyuan Ma, and Soumik Mandal. Reconciling modern machine-learning practice and the bias-variance trade-off. *Proceedings of the National Academy of Sciences*, 116(32):15849–15854, 2019. doi: 10.1073/pnas.1903070116.

Introduces the "double-descent" risk curve: beyond the interpolation threshold ($p > n$) the test error of minimum-norm interpolants can decrease again, but only under favourable geometric conditions not present in small-n physical-data problems.

Max Born. Zur Quantenmechanik der Stoßvorgänge. *Zeitschrift für Physik*, 37:863–867, 1926. doi: 10.1007/BF01397477. Introduces Born's rule: $|\psi|^2$ is the probability density for finding a particle. Born explicitly credits Einstein's treatment of the electromagnetic wave intensity as a photon probability density as the conceptual precedent. Nobel Prize in Physics, 1954.

Arthur H. Compton. A quantum theory of the scattering of X-rays by light elements. *Physical Review*, 21(5):483–502, 1923. doi: 10.1103/PhysRev.21.483. Compton scattering: independent confirmation of the photon. Nobel Prize in Physics, 1927.

Albert Einstein. Über einen die Erzeugung und Verwandlung des Lichtes betreffenden heuristischen Gesichtspunkt. *Annalen der Physik*, 17:132–148, 1905. doi: 10.1002/andp.19053220607. The "miraculous year" paper proposing the light quantum and deriving the linear V_0–ν relation. Nobel Prize in Physics, 1921.

Roy J. Glauber. The quantum theory of optical coherence. *Physical Review*, 130(6): 2529–2539, 1963. doi: 10.1103/PhysRev.130.2529. Introduces coherent states of the radiation field and the photon-number statistics that distinguish classical, coherent, and thermal light. Nobel Prize in Physics, 2005.

Robert Hanbury Brown and Richard Q. Twiss. Correlation between photons in two coherent beams of light. *Nature*, 177:27–29, 1956. doi: 10.1038/177027a0. Observes photon bunching (super-Poissonian coincidences) in thermal light; motivated Glauber's quantum coherence theory.

Gerald Holton. Centennial focus: Millikan's measurement of Planck's constant. *Physics (APS Physics Focus)*, 3:story 23, 1999. doi: 10.1103/physrevfocus.3.23. URL https://physics.aps.org/story/v3/st23.

Kurt Hornik, Maxwell Stinchcombe, and Halbert White. Multilayer feedforward networks are universal approximators. *Neural Networks*, 2(5):359–366, 1989. doi: 10.1016/0893-6080(89)90020-8. Proves that a single hidden layer with a non-constant squashing activation function can approximate any continuous function on a compact subset of $\mathbb{R}^n$.

Arthur Llewelyn Hughes. On the emission velocities of photo-electrons. *Philosophical Transactions of the Royal Society of London, Series A*, 212:205–226, 1913. doi: 10.1098/rsta.1913.0007. Submitted 1912. Three-frequency measurements for several metals; contact potentials uncontrolled; h values 3.55–5.85$\times 10^{-27}$ erg·s.

H. Jeff Kimble, Mario Dagenais, and Leonard Mandel. Photon antibunching in resonance fluorescence. *Physical Review Letters*, 39(11):691–695, 1977. doi: 10.1103/PhysRevLett.39.691. First experimental observation of photon antibunching (sub-Poissonian photon-number statistics) in the fluorescence of single sodium atoms; provides definitive evidence for quantisation of the radiation field, impossible in any classical or semiclassical model.

Willis E. Lamb and Marlan O. Scully. The photoelectric effect without photons. In French Physical Society, editor, *Polarisation, Matière et Rayonnement: Livre de Jubilé en l'Honneur du Professeur A. Kastler*, pages 363–369. Presses Universitaires de France, Paris, 1969. Shows that a semiclassical model (classical radiation field, quantised atom) reproduces Einstein's photoelectric stopping-potential equation; the photoelectric effect alone does not require quantisation of the radiation field.

Philipp Lenard. Über die lichtelektrische Wirkung. *Annalen der Physik*, 313(5):149–198, 1902. doi: 10.1002/andp.19023130510. Also cited as *Ann. Phys.* **8**:149–198 (neue Folge). Establishes that stopping potential is independent of intensity but depends on frequency. Nobel Prize in Physics, 1905.

Robert Andrews Millikan. A direct photoelectric determination of Planck's '*h*'. *Physical Review*, 7(3):355–388, 1916. doi: 10.1103/PhysRev.7.355. Definitive measurement. Full text at `archive.org`, identifier `sim_physical-review_1916-03_7_3`. Nobel Prize in Physics, 1923.

Abraham Pais. *'Subtle is the Lord . . . ': The Science and the Life of Albert Einstein*. Oxford University Press, Oxford, 1982. Authoritative scientific biography; see § 19 for the photoelectric effect.

Max Planck. Letter to R. W. Wood, 7 October 1931, 1931. Reproduced in A. Hermann, *The Genesis of Quantum Theory (1899–1913)*, MIT Press, Cambridge, MA, 1971, pp. 23–24. The key passage reads: "Briefly summarized, what I did can be described as simply an act of desperation. . . a theoretical interpretation had to be found at any cost, no matter how high.".

Owen Willans Richardson and Karl Taylor Compton. The photoelectric effect. *Philosophical Magazine*, 24:575–594, 1912. doi: 10.1080/14786441208637039. Measurements on eight metals; systematic underestimation of h by 10–45% attributed by Millikan to stray short-wavelength light and narrow frequency range.

Thomas J. Sargent. Sources of artificial intelligence. *Journal of Economic Dynamics and Control*, 172:104989, 2025.

Kai Siegbahn, Carl Nordling, Arne Fahlman, Roland Nordberg, Karl Hamrin, Johan Hedman, Gunnar Johansson, Tord Bergmark, Sven-Erik Karlsson, Ingvar Lindgren, and Bengt Lindberg. *ESCA: Atomic, Molecular and Solid State Structure Studied by Means of Electron Spectroscopy.* Almqvist & Wiksell, Uppsala, 1967. Introduces X-ray Photoelectron Spectroscopy (XPS/ESCA), applying Einstein's photoelectric equation at X-ray photon energies to measure core-electron binding energies element-specifically. Nobel Prize in Physics, 1981 (Kai Siegbahn).

Silviu-Marian Udrescu and Max Tegmark. AI Feynman: A physics-inspired method for symbolic regression. *Science Advances*, 6(16):eaay2631, 2020. doi: 10.1126/sciadv.aay2631. Demonstrates automated rediscovery of 100 physics equations, including photoelectric-type relations, from data; requires unit information and physical dimensions to match coefficients to fundamental constants.

Vladimir N. Vapnik. *Statistical Learning Theory.* Wiley, New York, 1998. Foundational treatment of hypothesis classes, VC dimension, and the bias-variance tradeoff.